

# Commerce XI Maths App

## Let's Remember

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 3 - Complex Numbers | Total questions: 9

## Question 1 - Let's Remember (i)

A number of the form  $a + ib$ , where  $a, b \in \mathbb{R}$  and  $i = \sqrt{-1}$ , is called

### Options

- A. a complex number
- B. an irrational number
- C. an imaginary number only
- D. a natural number

**Correct answer: a complex number**

### Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives a complex number.

## Question 2 - Let's Remember (ii)

If  $z_1 = a + ib$  and  $z_2 = c + id$ , then  $z_1 + z_2$  equals

### Options

- A.  $(a + c) + (b + d)i$
- B.  $(ac - bd) + (ad + bc)i$
- C.  $(a - c) + (b - d)i$
- D.  $(a + b) + (c + d)i$

**Correct answer:  $(a + c) + (b + d)i$**

### Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives  $(a + c) + (b + d)i$ .

## Question 3 - Let's Remember (iii)

If  $z_1 = a + ib$  and  $z_2 = c + id$ , then  $z_1 z_2$  equals

### Options

- A.  $(ac - bd) + (ad + bc)i$
- B.  $(ac + bd) + (ad - bc)i$
- C.  $(a + c) + (b + d)i$
- D.  $(ac - bd) + (ab + cd)i$

**Correct answer:  $(ac - bd) + (ad + bc)i$**

### Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives  $(ac - bd) + (ad + bc)i$ .

### Question 4 - Let's Remember (iv)

For a positive integer  $k$ ,  $i^{4k}$  equals

**Options**

- A. 1
- B. -1
- C.  $i$
- D.  $-i$

**Correct answer: 1**

**Explanation and steps**

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives 1.

### Question 5 - Let's Remember (v)

For a positive integer  $k$ ,  $i^{4k+1}$  equals

**Options**

- A.  $i$
- B. 1
- C. -1
- D.  $-i$

**Correct answer:  $i$**

**Explanation and steps**

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives  $i$ .

### Question 6 - Let's Remember (vi)

For a positive integer  $k$ ,  $i^{4k+2}$  equals

**Options**

- A. -1
- B. 1
- C.  $i$
- D.  $-i$

**Correct answer: -1**

**Explanation and steps**

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives -1.

## Question 7 - Let's Remember (vii)

For a positive integer  $k$ ,  $i^{4k+3}$  equals

### Options

- A.  $-i$
- B.  $i$
- C.  $1$
- D.  $-1$

**Correct answer:**  $-i$

### Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives  $-i$ .

## Question 8 - Let's Remember (viii)

The conjugate of  $z = a + ib$  is

### Options

- A.  $\bar{z} = a - ib$
- B.  $\bar{z} = -a + ib$
- C.  $\bar{z} = a + ib$
- D.  $\bar{z} = b + ia$

**Correct answer:**  $\bar{z} = a - ib$

### Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives  $\bar{z} = a - ib$ .

## Question 9 - Let's Remember (ix)

The cube roots of unity are denoted by

### Options

- A.  $1, w, w^2$  or  $1, w, \bar{w}$
- B.  $1, i, -i$
- C.  $0, w, w^2$
- D.  $1, w^2, w^3$

**Correct answer:**  $1, w, w^2$  or  $1, w, \bar{w}$

### Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives  $1, w, w^2$  or  $1, w, \bar{w}$ .